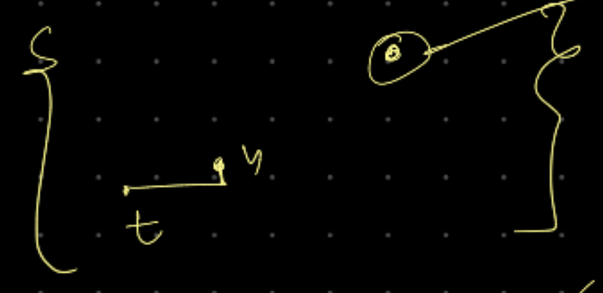
Snake Game



food item

snake's head → { U D L R }

make the snake longer → more points.

if I (snake), eat the (F.I), I get longer → points.

1) If I touch myself → game over.
2) If I touch the boundry game is over.

Snake itself :→

movement → adding to the first
→ removing the last

→ f.I is consumed → adding to the Begin but wont remove from end.

Double Ending Queue ⇒
↳ O(1) → { Deque }

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maintain scores

fooditems → N ⇒ (+1)
→ S ⇒ (+5)

score observer
⇓
subscribed to eating f.I

States :→ IGameState → move, pause, resume

→ Running, Paused, GameOver

functions of the state contract

implements ✓
extends ✓

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factory pattern :→

N.f. Item (+1), B.f. item (+5) - - - - ▷ IFoodItem
↳ position
points.

Poison

factory method

abstract food Creator
→ abstract Creat food
public → spawn at (position) { — } ⇒

N f creator

Bonus food Creator

Strategy :→

I food Spawn Strategy

Random
like 20% B.f sp
80% → N.f. spawn

after N s pawns
after 6 N.f → 1 B.f

Template Design Pattern :→

Levels → 1 → boundry Death
→ 2 → No Boundry Death → bounce back